SC 07 · CAPSTONE

Tune it, test it, report it

Numbers against a written specification, the plots that show them, and an honest two pages.

Two tracks

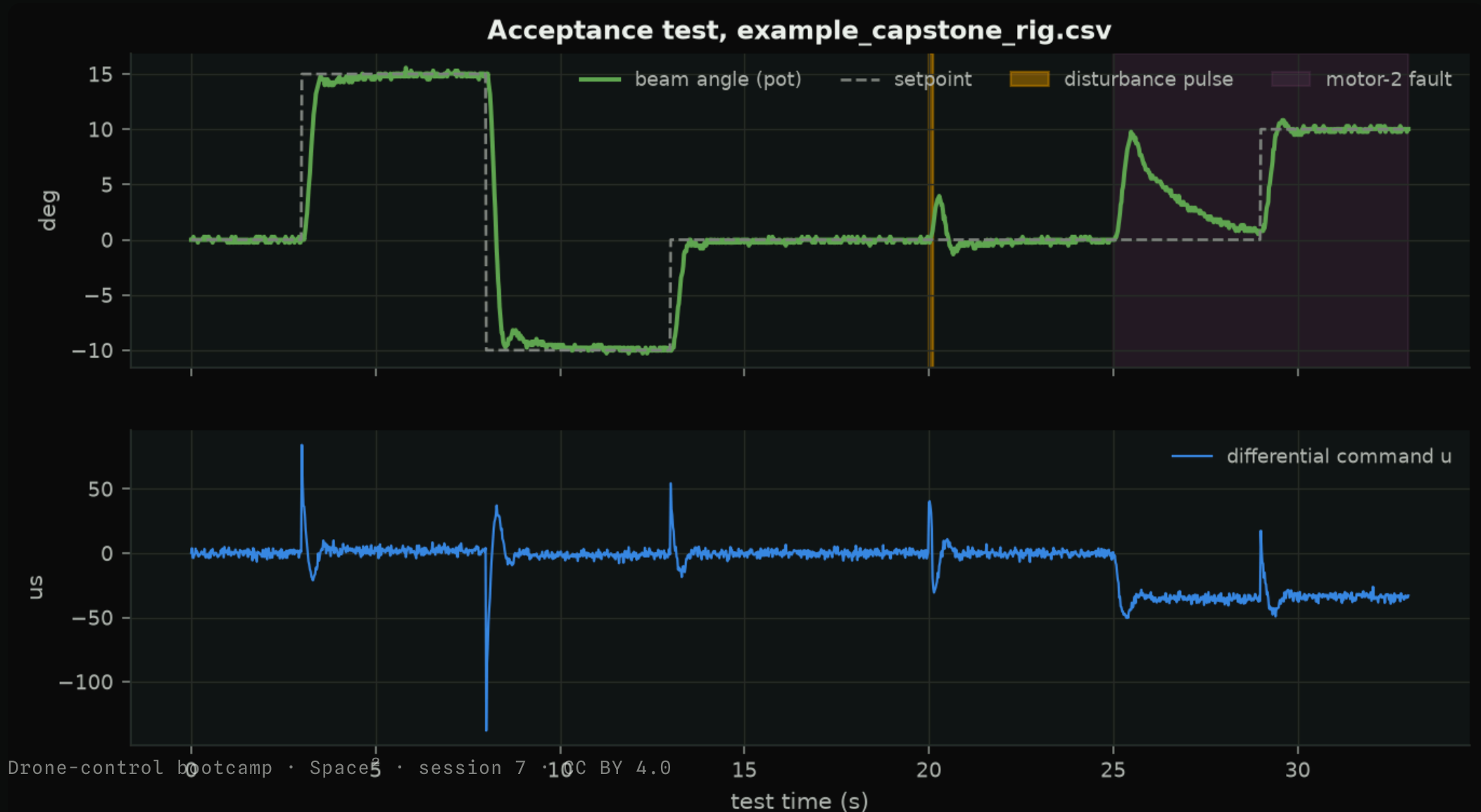
R · the rig

`s07_capstone`, command `test`: steps, a disturbance pulse, an injected motor fault. Worst of three runs against `RIG_SPEC`.

S · simulation

Quadrotor course with 50 % loss on rotor 1 at t = 9 s. Replace the oracle with **your** detector and estimator.

The acceptance test



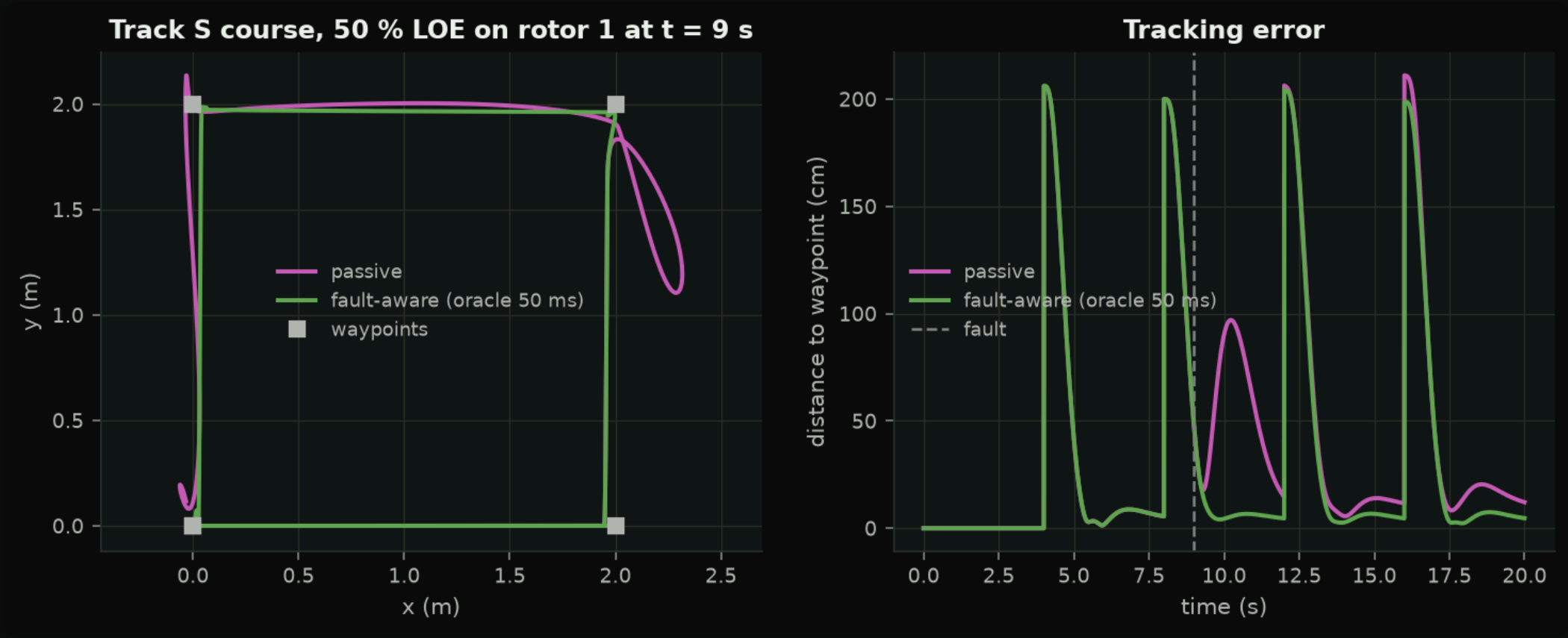
CC BY 4.0

The specification (track R)

line	limit
+15° step overshoot	≤ 20 %
settling into ±1°	≤ 2.0 s
hold error	≤ 1.0°
disturbance peak / recovery	≤ 8° / ≤ 1.5 s
error with the fault	≤ 1.5°
command chatter (RMS)	≤ 25 µs

Drone-control bootcamp · Space² · session 7 · CC BY 4.0

Track S: the course



Drone-control bootcamp · Space² · session 7 · CC BY 4.0

How to work

1. model first: your measured parameters, two seeds, margin
2. hardware gently: half the gains, then 25 % steps
3. test the way you are scored, log everything
4. explain the gap between model and hardware

An honest, measured failure scores better than an unexplained pass.

Drone-control bootcamp · Space² · session 7 · CC BY 4.0

Tonight

time	
0:00	stand-up
0:10	studio
1:50	acceptance tests
2:30	demos: one plot, one number, one surprise
2:50	quiz 7, feedback

Write-up due in one week.

Drone-control bootcamp · Space² · session 7 · CC BY 4.0